InCELLia Participant Profile

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organization (SMEs, NGOs , patient organizations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	INCELLIA IKE
Short name / acronym used in projects	InCELLia
Participant Identification Code (PIC), if available	888772686
Website(s)	https://incellia.com/
Country	Greece
Organisation type	☐ Academic ☑ Non-academic ☐ Hospital / clinic ☑ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Dimitriou Gounari 96, Amarousion, 15125, Athens Greece
Department(s), institute(s), research group(s) or unit(s) that would be involved	InCELLia research team in mathematical modelling, biomedical data analysis and experimental model integration, led by Dr Georgios Lolas.
Additional institutional resources relevant to collaboration	Computational models and analysis workflows; expertise in transparent fish models, including Danionella cerebrum; collaborations linking modelling with biological experiments. Access to specific partner facilities or datasets would require a separate agreement.

Main contact person for this profile	Dr Georgios Lolas, Founder and Legal Representative; applied mathematician specialising in developing mathematical models for biological processes (with special focus on cancer); scientific contact for InCELLia.
Main contact email	glolas@incellia.com
Other contact person(s), if relevant	N/A

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☑ **Inflammation / biomarkers / neuroprotection**	☑ **Drug delivery / formulations / nanomedicine**
☑ **Drug discovery / screening / repurposing**	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☑ **AI-assisted diagnosis / software / digital tools**
☐ Clinical trials / translational validation	☑ Other: mechanistic disease modelling; Bayesian inference; longitudinal data analysis; transparent fish models (*danionella cerebrum*). Eye-health applications would be developed with specialist partners.

Free Keywords for internal searchable database (please provide up to 10)	Mechanistic modelling; machine learning; Bayesian inference; Danionella cerebrum; zebrafish; drug-response prediction; biomarkers; longitudinal imaging; drug-delivery modelling; experimental design.
Geographies of interest for exchanges (optional)	International collaboration, including IOBA in Spain. Destinations for any outgoing secondments would be agreed according to scientific fit and staff availability.

Overview of your research / innovation focus
InCELLia is a Greek research SME combining mathematical modelling, machine learning and experimental model integration to study disease mechanisms, toxicity and treatment response. Its main application area is oncology, including primary tumour growth, metastatic progression and the evaluation of therapeutic candidates. The team develops ordinary and partial differential equation models and structured population models, fits them to experimental observations, and uses Bayesian and statistical methods to quantify uncertainty and biological variability. InCELLia connects these computational approaches with research using transparent fish models, including Danionella cerebrum, and with longitudinal measurements from collaborating experimental teams. Its contribution to the ALADDIN project includes mathematical models for therapeutic target identification and preclinical validation of nanobodies. For an eye-health consortium, InCELLia could adapt these methods to questions selected with ophthalmology partners, such as inflammation, tissue damage and repair, drug exposure and treatment response. It could integrate imaging, biomarker and molecular measurements to compare biological hypotheses, identify useful measurements and support experimental planning. These would be jointly developed research applications, with disease-specific validation provided through the consortium.

Why your team could be relevant to a translational eye-health and vision consortium
InCELLia could help turn experimental and clinical observations into quantitative explanations and testable predictions. Models of inflammation, cell survival and tissue responses could be adapted to ocular disease, while transport and exposure models could support the comparison of drug-delivery strategies. Machine learning could help assess candidate biomarkers and combine imaging with molecular measurements, subject to sufficient data and independent validation. Its experience with transparent fish models could also support feasibility studies for longitudinal measurements and treatment screening in collaboration with eye-research specialists. The main added value is a repeated exchange between modelling and experiments: observations inform the model, predictions guide the next experiment, and new results test those predictions.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
Mechanistic modelling of biological dynamics using ordinary and partial differential equations, structured population models and spatial models. Potential applications include disease progression, inflammatory signalling, cell interactions and treatment response. Bayesian inference, parameter estimation, mixed-effects analysis, model comparison, sensitivity analysis and uncertainty quantification. These methods help assess which conclusions are supported by the available measurements and which additional data would be useful. Machine learning and statistical analysis of biomarker, molecular, imaging-derived and longitudinal data, including feature selection and predictive model assessment. Integration of mechanistic models with data-driven approaches where this improves interpretation or prediction. Drug-exposure, transport and response modelling, including physiologically based pharmacokinetic and population balance approaches where appropriate. Design and interpretation of studies using transparent fish models, with model predictions compared against experimental observations.

Infrastructure, equipment, facilities or datasets relevant to consortium work
InCELLia can contribute computational models, numerical simulation and data-analysis workflows, together with expertise in Danionella cerebrum research. Specific experimental resources, imaging access and study datasets would be defined with the participating partners. Clinical cohorts, ocular samples and specialist ophthalmic facilities would need to be provided through those collaborations. This profile does not commit external facilities or datasets to the consortium.

Training capacity and knowledge-transfer potential
InCELLia **cannot** host incoming staff. It can contribute knowledge transfer through agreed outgoing secondments, workshops at partner institutions and remote mentoring. Topics include translating biological questions into mathematical models, fitting models to data, evaluating uncertainty, using machine learning responsibly, and linking experimental design with computational predictions. Training can use shared case studies and reproducible analysis workflows.

What your team would hope to gain from participating
InCELLia would seek ophthalmology expertise, disease-specific experimental models, longitudinal ocular imaging and biomarker data, and access to relevant validation studies through collaboration. The team would benefit from learning which outcomes matter to patients and clinicians, how ocular therapies are assessed, and how processes like angiogenesis (that are similar to cancer development) contribute to treatment response/success and how its modelling and experimental approaches can be adapted to these needs. Joint publications and durable academic-industry research partnerships are also of interest.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Dr Georgios Lolas	Founder and Legal Representative; scientific lead	InCELLia research team	Researcher	Experienced researcher	glolas@incellia.com Orcid: 0000-0002-2236-5209
Aris Chatzichristos	Physics modeler, Data analyst	InCELLia research team	Researcher	Junior Researcher	

Main disciplines represented in your team	Applied mathematics, mathematical biology, computational oncology, statistics, machine learning and biomedical data integration, linked with experimental biology through collaborative research.
Experience in EU / international / multi-partner collaborative projects	Beneficiary in ALADDIN, a Horizon Europe EIC Pathfinder project, contributing mathematical modelling for target identification and preclinical validation within an international multidisciplinary consortium.
Experience with clinical, translational or patient-facing research (if relevant)	☑ Yes ☐ No ☐ Not applicable Translational and preclinical research through ALADDIN and model-experiment integration. Any patient-facing work would be undertaken by clinical partners.

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

1. Gavriliadis P, Lolas G, Matsoukas T. Age and generation-based model of metastatic cancer: From micrometastases to macrometastases. Journal of Theoretical Biology 623 (2026), 112392. Demonstrates expertise in structured population modelling and the quantitative interpretation of disease progression. https://doi.org/10.1016/j.jtbi.2026.112392
2. Kavidopoulou A.,…., Lolas G. (2022), AI and Big Data for Drug Discovery. In: Sakly, H., Yeom, K., Halabi, S., Said, M., Seekins, J., Tagina, M. (eds) Trends of Artificial Intelligence and Big Data for E-Health. Integrated Science, vol 9. Springer, Cham. A review on the latest advancements on AI and Big Data for Drug Discovery
3. Dlamini, Z., Syrigos K.N., Evangelou G., Vamvakaris I., Lolas G. (2022). AI and Big Data for Cancer Segmentation, Detection and Prevention. In: Sakly, H., Yeom, K., Halabi, S., Said, M., Seekins, J., Tagina, M. (eds) Trends of Artificial Intelligence and Big Data for E-Health. Integrated Science, vol 9. Springer, Cham. A review on cancer segmentation using AI approaches.
4. Ali Z, ... G, Lolas G, Syrigos KN, Jensen LD. Zebrafish patient-derived xenograft models predict lymph node involvement and treatment outcome in non-small cell lung cancer. J Exp Clin Cancer Res. 2022 Feb 9;41(1):58. doi: 10.1186/s13046-022-02280-x. A collaboration effort between NKUA, INCELLIA and LiU to justify that ZTX platform can provide fast accurate and clinical treatment outcomes using NSCLC tumor biopsies (patient derived) send from Greece.
5. Lolas G, Bianchi A, Syrigos KN. Tumour-induced neoneurogenesis and perineural tumour growth: a mathematical approach. Sci Rep. 2016 Feb 10;6:20684. doi: 10.1038/srep20684. PMID: 26861829; PMCID: PMC4748234.

Contribution to ALADDIN modelling workflows for therapeutic target identification and preclinical validation. The public consortium profile documents InCELLia involvement in mathematical modelling across these activities. https://aladdin-project.eu/consortium/

Up to 5 relevant previous projects, grants, networks or collaborations

1. ALADDIN — Accelerated Discovery Nanobody Platform. Funding: European Union, Horizon Europe, EIC Pathfinder Open. Grant Agreement 101130574. Duration: 1 January 2024 to 31 December 2027. Role: beneficiary/partner in an international consortium. InCELLia contributes mathematical models for target identification and preclinical validation, linking computational analysis with experimental evaluation of therapeutic candidates. https://cordis.europa.eu/project/id/101130574
2. ''Emerging mechanisms of tumor-lymphangiogenesis in metastasis: bridging the gaps with a Systems Medicine approach for improve preventive strategies". Our ultimate goal is to develop mathematical and computational models that highlight the crucial role of the lymphatic system in metastasis. In collaboration with National Kapodistrian University of Athens (NKUA).
3. Detection of lung cancer cells on Whole Slide Images using Deep Convolution Neural Networks" We assist the clinical team of the Oncology Unit of NKUA to develop a deeplearning- based CAD method for lung biopsy tissue classification, to optimise and assist histopathological imaging outcomes. In collaboration with UOA.
4. Slow-dividing E.coli competes with cancer cell lines for Arginin"
   Highlight: identify, fey extracellular nutrients can be exploited in tumor therapy: depletion of nutrients, degradation of enzymes, and inhibition of nutrient uptake and utilization. In collaboration with UOA.
5. "Development of an Artificial Intelligence Predictive model for Metastasis based on patient derived Zebrafish xenografting imaging data", a joined work with INCELLiA, UOA and LiU. The aim of this joined work is to develop a predictive model based on patient derived xenograft imaging data in order to identify the metastatic propensity and aggressiveness of the initial primary tumor. The tumor under consideration is NSCLC.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
Shareable know-how includes mathematical model formulations, calibration methods and computational analysis workflows for biological systems and therapeutic response. Access to existing code, models and project data would be agreed for each collaboration, with background intellectual property and confidentiality clearly defined. Any ophthalmology-specific application would require dedicated development and validation.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organization's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organization/team second staff?	☐ Yes ☑ Maybe ☐ No capacity to send staff on secondment Outgoing secondments can be discussed. Staff numbers, duration and total person-months are not yet committed.
Could your organization second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles Research staff in modelling and data analysis are the envisaged profile. Final staff categories remain to be agreed.
Could your organization host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only ☐ Both ☑ No capacity to host InCELLia cannot host incoming staff. Incoming secondment allocation: 0 staff and 0 person-months.
Administrative / HR / legal considerations that may affect mobility	Any outgoing mobility depends on staff availability, continuity of company activities, an agreed research and training plan, and suitable arrangements at the receiving organisation. Employment, insurance, access, confidentiality and intellectual-property matters would be settled before an exchange. No incoming secondments should be assigned to InCELLia.

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Mechanistic models that explain disease dynamics and treatment response; Bayesian and machine-learning methods that combine biological data with quantified uncertainty; experience connecting computational predictions with longitudinal observations in transparent fish models.

Capabilities or expertise you would especially like to access through the consortium
Ophthalmology and ocular biology teams; specialists in inflammation and neuroprotection; drug-delivery and nanomedicine groups; ocular imaging and biomarker datasets; and experimental or clinical partners able to evaluate jointly developed models against relevant outcomes.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	☑Yes ☐ No

☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
InCELLia cannot host incoming staff. Its contribution should therefore be planned around research collaboration, agreed outgoing secondments and knowledge-transfer activities delivered with partner organisations. Any outgoing staff allocation remains non-binding until the scientific tasks, timing and resources are agreed. The proposed ocular applications build on transferable modelling, data-analysis and experimental expertise. Specific disease models, datasets, experimental responsibilities and validation endpoints should be defined jointly with IOBA and the relevant partners. Access to background models, code and data, as well as ownership and use of jointly developed results, should be agreed before the work starts.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☐ **Understood**

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Date completed: 17 September 2026

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